SdkApi Guidance document

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Overview

APIs supported by motherboards running Android 10 and above, along with their corresponding descriptions.

Target Audience

This document is mainly intended for the following engineers:

- Software Development Engineers
- Technical Support Engineers
- Sales Engineers

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API Detailed Description

API Import Method

Import aar package into Android Studio

1、Copy the araar package [SdkApiJar.aar] to the project directory /app/libs.



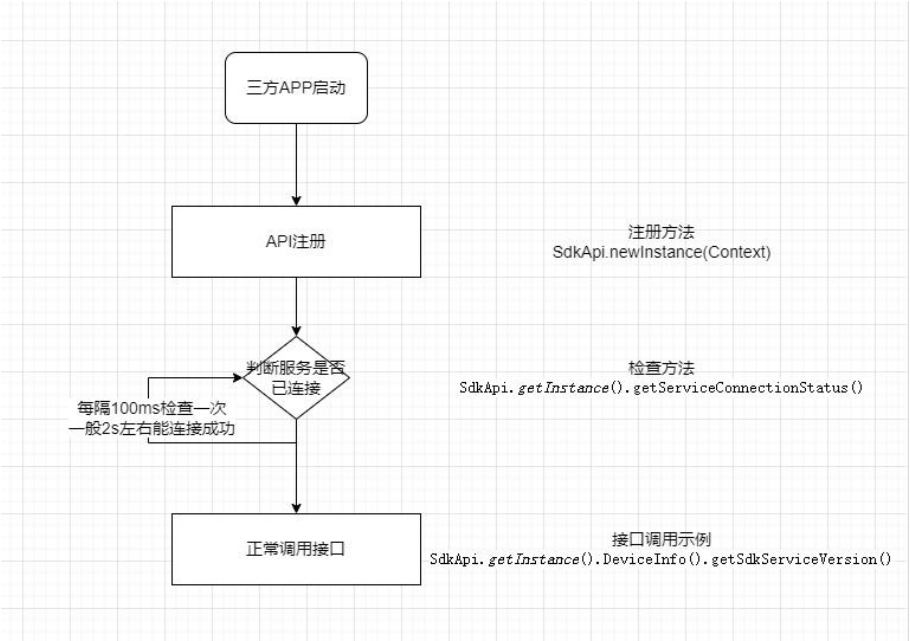
2、Add references to build.gradle files in the app.



3、Register SdkApi.

4、Use the functions of the SdkApi-related class interface.

API Registration Flowchart



API Module Classification

module	Class Name	meaning
facility information	ApiDeviceInfo	Device Basic Info Interface
display control	ApiDisplay	Display relevant control interfaces
System control	ApiSystemCtrl	System control interface
Schedule	ApiTimerOnoff	Timer interface
Dog Settings	ApiWatchdog	Dog survival-related interface
Time Settings	ApiSystemTime	System time-related interfaces
hardware control	ApiHardware	Hardware control interface
Other Settings	ApiOther	Other related interfaces
Scale set up	ApiScale	Touchpoint chip control interface
General Settings	ApiCommonEvent	General Function Interface

1. Initialize API

1.1 Bind SdkApi

Interface name:
void newInstance(Context context)

Parameter：
--context: context

Returned value：
-- None

Code example:

```
    SdkApi.newInstance(this);
```

Remarks：
-- None

1.2 Obtain the SdkApi instance

Interface name:
SdkApi getInstance()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	class	SdkApi living example

Code example:

```
    SdkApi api = SdkApi.getInstance();
```

Remarks：
-- None

1.3 Get the SDK service connection status

Interface name:
boolean getServiceConnectionStatus()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true: Connection successful false: Connection failed

Code example:

```
boolean ret = SdkApi.getInstance().getServiceConnectionStatus();
```

Remarks：
After the--newInstance function completes, wait approximately 2 seconds for the service to connect successfully.

1.4 Unbind SDK API

Interface name:
void unBindApiSdkService()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	void	

Code example:

```
SdkApi.getInstance().unBindApiSdkService();
```

Remarks：
--After unbinding, you need to rebind the SDK API

2. Facility information API

--Interface call method: SdkApi.getInstance(). DeviceInfo(). Interface

2.1 Get the API aar library version

Interface name:
String getApiVersion()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	aar library version

Code example:

```
String version =
SdkApi.getInstance().DeviceInfo().getApiVersion();

Version: YYMMdd.hhmm
Instance： 250813.0759
```

Remarks：
-- None

2.2 Get API SDK service version

Interface name:
String getApiSdkServiceVersion()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	SDK service version

Code example:

```
String version =
SdkApi.getInstance().DeviceInfo().getApiSdkServiceVersion();

Version: V2.YYMMdd.hhmm ss
Example: V2.20 250813.0759 00
```

Remarks：
-- None

2.3 Get the motherboard system version

Interface name:
String getFirmwareVersion()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	System version

Code example:

```
String version =
SdkApi.getInstance().DeviceInfo().getFirmwareVersion();

Example: EC3576_RK3576_Android14_A_20250813.143037
```

Remarks：
-- None

2.4 Obtain the mainboards main chip name

Interface name:
String getChipName()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	Main chip name

Code example:

```
String chip = SdkApi.getInstance().DeviceInfo().getChipName()
```

Instance： rk3576

Remarks：
-- None

2.5 Get the motherboard serial number

Interface name:
String getSerialNumber()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	Mainboard unique serial number

Code example:

```
String sn = SdkApi.getInstance().DeviceInfo().getSerialNumber()
```

Instance： 12345678

Remarks：
--The data length is usually 8 or 21 bits

2.6 Get motherboard memory size

Interface name:
String getDDRSize()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	Total memory size

Code example:

```
String ddrSize = SdkApi.getInstance().DeviceInfo().getDDRSize()
```

Instance： 4GB

Remarks：
-- None

2.7 Get motherboard storage size

Interface name:
String getEMMCSize()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	Total storage size

Code example:

```
String ddrSize = SdkApi.getInstance().DeviceInfo().getEMMCSize()
```

Instance： 32GB

Remarks：
-- None

2.8 Get MCU version

Interface name:
void getMcuVersion(ResultCallback<ResultInfo<String>> cb)

Parameter：

Parameter name	Type	Explain
syntony	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	String	MCU version number

Code example:

```
SdkApi.getInstance().DeviceInfo().getMcuVersion(
    new ResultCallback<ResultInfo<String>>() {
        @Override
        public void onMsg(String val) {

        }
});
```

Remarks：
--Based on the actual motherboard, some motherboards do not have an MCU and do not support this interface.

2.9 Set system screen density

Interface name:
void setLcdDensity(int density)

Parameter：

Parameter name	Type	Explain
syntony	int	Screen density Value range: 100-480 Default: 160

Returned value：
-- None

Code example:

```
SdkApi.getInstance().DeviceInfo().setLcdDensity ( density);
```

Remarks：
Some systems need to restart to take effect

2.10 Get system screen density

Interface name:
int getLcdDensity()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	int	Screen density

Code example:

```
int density = SdkApi.getInstance().DeviceInfo().getLcdDensity()
```

Remarks：
-- None

2.11 Set system prop properties

Interface name:
void setProperty(String key, String value)

Parameter：

Parameter name	Type	Explain
key	String	Prop key name
value	String	value of the prop key

Returned value：
-- None

Code example:

```
SdkApi.getInstance().DeviceInfo().setProperty("
persist.sys.timezone" , "Asia/Shanghai" )
```

Remarks：
--ro attribute setting is not supported. Due to selinux permission issues in Android 10/11, it is recommended to start with vendor.

2.12 Get the system prop attribute

Interface name:
String getProperty(String key, String default)

Parameter：

Parameter name	Type	Explain
key	String	Prop key name
default	String	Prop key default value

Returned value：

Parameter name	Type	Explain
parameter values	String	parameter value corresponding to the system prop attribute

Code example:

```
String prop = SdkApi.getInstance().DeviceInfo().getProperty(
" persist.sys.timezone" , "Asia/Shanghai" )
```

Remarks：
-- None

2.13 Get the motherboards Ethernet MAC address

Interface name:
String getEthMac()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	Ethernet MAC address, format: "0C63FC112233"

Code example:

```
String mac =  SdkApi.getInstance().DeviceInfo().getEthMac()
```

Remarks :

When a board has multiple MAC addresses, the results are separated by @. For example: 0C63FC112233@0C63FC445566
--> Previous Mac: Ethernet 0
----> The latter Mac: Ethernet 1

2.14 Get Motherboard Available Memory Size

Interface name:
`String getDDRAvailableSize()`

Parameter :
-- None

Returned value :

Parameter name	Type	Explain
Returned value	String	Total Motherboard Memory Size

Code example:

```
String ddrSize =
SdkApi.getInstance().DeviceInfo().getDDRAvailableSize()


Example: 2.6GB
```

Remarks：
-- None

2.15 Get Motherboard Available Storage Size

Interface name:
String getEMMCAvailableSize()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	Total Motherboard Storage Size

Code example:

```
String ddrSize =
SdkApi.getInstance().DeviceInfo().getEMMCAvailableSize()

Example: 32.0GB
```

Remarks：
-- None

3. Show relevant API

--Interface call method: SdkApi.getInstance(). Display(). Interface

--screen display interface ID includes

ID	Define in.aar	Interface value	Explain
ALL	ApiTool.Screen.ALL	0	Set up all display interfaces on the motherboard
LVDS	ApiTool.Screen.LVDS	1	LVDS interface defined in motherboard specifications
LVDS2	ApiTool.Screen.LVDS2	2	LVDS2 interface defined in motherboard specifications
eDP	ApiTool.Screen.eDP	3	The eDP interface defined in the motherboard specifications
MIPI	ApiTool.Screen.MIPI	4	The MIPI interface defined in the motherboard specifications
VB1	ApiTool.Screen.VB1	5	The VB1 interface defined in the motherboard specifications
HDMI	ApiTool.Screen.HDMI	6	HDMI interface defined in motherboard specifications
DP	ApiTool.Screen.DP	7	DP interface defined in motherboard specifications

3.1 Set screen orientation

Interface name:
boolean setScreenRotate(int screen, int rotate)

Parameter：

Parameter name	Type	Explain

screen	int	Display interface ID
rotate	int	Direction: 0/90/180/270

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display().setScreenRotate(
ApiTool.Screen.ALL, 90)
```

Remarks：
Restart the system to take effect

3.2 Get screen orientation

Interface name:
void getScreenRotate(int screen, ResultCallback<ResultInfo<Integer>> cb)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	int	Error code: 0/90/180/270

Code example:

```
SdkApi.getInstance().Display().getScreenRotate(ApiTool.Screen.ALL
,
    new ResultCallback<ResultInfo<Integer>>() {
        @Override
        public void onMsg(Integer val) {

        }
})
```

Remarks：
If the motherboard has dual outputs, setting screen to all returns the current main screen status.

3.3 Set screen switch

Interface name:
boolean setScreenOnoff(int screen, boolean onoff)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
onoff	boolean	true： open false： close

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display().setScreenOnoff(
ApiTool.Screen.ALL, true)
```

Remarks：
-- None

3.4 Get screen switch

Interface name:
void getScreenOnoff(int screen, ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	String	true： open false： close

Code example:

```
SdkApi.getInstance().Display().getScreenOnoff(ApiTool.Screen.ALL,
    new ResultCallback<ResultInfo<Boolean>>() {
        @Override
        public void onMsg(Boolean val) {

        }
})
```

Remarks：
If the motherboard has dual outputs, setting screen to all returns the current main screen status.

3.5 Set screen brightness

Interface name:
boolean setBrightness(int screen, int brightness)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
brightness	int	screen intensity Range: 0-100

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display().setBrightness(
ApiTool.Screen.ALL, 100)
```

Remarks：
-- None

3.6 Get screen brightness

Interface name:
void getBrightness(int screen, ResultCallback<ResultInfo<Integer>> cb)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	int	Screen brightness: 0-100

Code example:

```
SdkApi.getInstance().Display().getBrightness(ApiTool.Screen.ALL,
    new ResultCallback<ResultInfo<Integer>>() {
    @Override
        public void onMsg(Integer val) {

        }
})
```

Remarks：
If the motherboard has dual outputs, setting screen to all returns the current main screen status.

3.7 Set backlight inversion

Interface name:
boolean setBacklightInverse(int screen, boolean inverse)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
inverse	boolean	Adjust backlight reverse. Default: false false: backlight voltage 0-3.3/5V or PWM duty cycle 0-100 variation True: backlight voltage 3.3/5-0V or PWM duty cycle 100-0 variation

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display().setBacklightInverse(
ApiTool.Screen.ALL, false)
```

Remarks：
--Use the new interface setBacklightParams

3.8 Get backlight reversed

Interface name:
void getBacklightInverse(int screen, ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	true： open false： close

Code example:

```
SdkApi.getInstance().Display().getBacklightInverse(
    ApiTool.Screen.ALL,
    new ResultCallback<ResultInfo<Boolean>>() {
        @Override
        public void onMsg(Boolean val) {

        }
})
```

Remarks：
If the motherboard has dual outputs, setting screen to all returns the current main screen status.
--Use the new getBacklightParams interface

3.9 Set the maximum and minimum PWM backlight values

Interface name:
boolean setBacklightPWM(int screen, int min, int max)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
min	int	Range: 0-100
max	int	Range: 0-100

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display().setBacklightPWM(
ApiTool.Screen.ALL, 0, 0)
```

Remarks：
The setting does not take effect when the minimum value is greater than the maximum value.
--Use the new interface setBacklightParams

3.10 Obtain the maximum and minimum values of backlight PWM

Interface name:
void getBacklightPWM(int screen, ResultCallback<ResultInfo<String>> cb)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	String	Error: string --The previous value is the minimum --The maximum value follows Example: "10,100" indicates a minimum value of 10 and a maximum value of 100.

Code example:

```
SdkApi.getInstance().Display().getBacklightPWM(ApiTool.Screen.ALL
,
    new ResultCallback<ResultInfo<String>>() {
        @Override
        public void onMsg(String val) {

        }
})
```

Remarks：
--Use the new getBacklightParams interface

3.11 Set screen resolution

Interface name:
boolean setResolution(int screen, String resolution)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
resolution	String	Resolution format: width x height Note: When the screen is HDMI, add "@frame rate". for Instance： lvds：“1920x1980” hdmi：“3840x2160@25”

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display().setResolution(
ApiTool.Screen.LVDS, "1920x1080")
```

Remarks：
--Restart to take effect

3.12 Get screen resolution

Interface name:
void getResolution(int screen, ResultCallback<ResultInfo<String>> cb)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	String	Return value: resolution Format: "1920x1080@60"

Code example:

```
SdkApi.getInstance().Display().getResolution(ApiTool.Screen.ALL,
    new ResultCallback<ResultInfo<String>>() {
        @Override
        public void onMsg(String val) {

        }
})
```

Remarks：
If the motherboard has dual outputs, setting screen to all returns the current main screen status.

3.13 Set LVDS mode

Interface name:
boolean setLvdsMode(int screen, int mode)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID support -- LVDS -- LVDS2 --ALL (updates only the LVDS interface) Other interfaces are not supported
mode	int	0：vesa 1：jeida

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display().setLvdsMode(
ApiTool.Screen.ALL, 0)
```

Note:
--None

3.14 Get LVDS mode

Interface name:
void getLvdsMode(int screen, ResultCallback<ResultInfo<Integer>> cb)

Parameters:

Parameter name	Type	Description
screen	int	Display interface ID Support -- LVDS -- LVDS2 --ALL (updates only the LVDS interface) Other interfaces are not supported
cb	class	ResultCallback class

Return value:

Parameter name	Type	Description
Return parameter val	int	Error handler: 0：vesa 1：jeida

Code example:

```
SdkApi.getInstance().Display().g etLvdsMode(ApiTool.Screen.ALL,
    new ResultCallback<ResultInfo< Integer>>() {
        @Override
        public void onMsg( Integer val) {

        }
})
```

Note:
If the motherboard has dual outputs, setting screen to all returns the current main screen status.

3.15 Configure LVDS Even-ODD switching

Interface name:
boolean setLvdsSwap(int screen, int swap)

Parameters:

Parameter name	Type	Description
screen	int	Display interface ID Support -- LVDS -- LVDS2 --ALL (updates only the LVDS interface) Other interfaces are not supported
swap	int	0：no swap 1：swap

Return value:

Parameter name	Type	Description
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display().setLvdsMode(
ApiTool.Screen.ALL, 0)
```

Remarks：
-- None

3.16 Obtain LVDS Even-ODD switching

Interface name:
void getLvdsSwap(int screen, ResultCallback<ResultInfo<Integer>> cb)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID support -- LVDS -- LVDS2 --ALL (updates only the LVDS interface) Other interfaces are not supported
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	int	syntony val： 0：no swap 1：swap

Code example:

```
SdkApi.getInstance().Display().getLvdsSwap(ApiTool.Screen.ALL,
    new ResultCallback<ResultInfo<Integer>>() {
        @Override
        public void onMsg(Integer val) {

        }
})
```

Remarks：
If the motherboard has dual outputs, setting screen to all returns the current main screen status.

3.17 Set LVDS channel

Interface name:
boolean setLvdsChannel(int screen, int ch)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID support -- LVDS -- LVDS2 --ALL (updates only the LVDS interface) Other interfaces are not supported
ch	int	0：single 1：dual

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display().setLvdsChannel(
ApiTool.Screen.ALL, 0)
```

Remarks：
-- None

3.18 Get LVDS channel

Interface name:
void getLvdsChannel(int screen, ResultCallback<ResultInfo<Integer>> cb)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID support -- LVDS -- LVDS2 --ALL (updates only the LVDS interface) Other interfaces are not supported
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	int	syntony val： 0：single 1：dual

Code example:

```
SdkApi.getInstance().Display().getLvdsChannel(ApiTool.Screen.ALL,
    new ResultCallback<ResultInfo<Integer>>() {
        @Override
        public void onMsg(Integer val) {

        }
});
```

Remarks：
If the motherboard has dual outputs, setting screen to all returns the current main screen status.

3.19 Get supported resolutions

Interface name:
String getSupportedResolutionsList(int screen)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID

Returned value：

Parameter name	Type	Explain
Return parameter val	String	Supported resolution list string in compliance with：“1920x1080@60,1366x768@60” If the support list is empty, return "none"

Code example:

```
String resolutionList =  SdkApi.getInstance().Display().get
SupportedResolutionsList (ApiTool.Screen.LVDS )


--Example: 1920x1080@60,1366x768@60
```

Remarks：
When ID is ALL, only LVDS interface data is returned.

3.20 Set Custom Resolution Timing

Interface name:
boolean setCustomTiming(CustomTiming timing)

Parameter：

Parameter name	Type	Explain
timing	CustomTiming	Custom Resolution Timing

structure declaration

```
Class CustomTiming{
int Type; //Screen index in ApiTool.TimingType
  int d_clk;
  int h_active;
  int v_active;
  int h_bp;
    int h_fp;
  int h_sync;
  int v_bp;
  int v_fp;
  int v_sync;
}

--field value definition
  >> 0: LVDS
  >> 1: OTHER2LVDS
  >> 2: MIPI
  >> 3: EDP
  >> 4: LVDS2
```

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display.setCustomTiming(new
CustomTiming(...));
```

Remarks：
Restart is required after import

3.21 Set custom MIPI commands

Interface name:
boolean setCustomMipiCmds(int[] cmdsBuf, int cmdsLen)

Parameter：

Parameter name	Type	Explain
cmdsBuf	int[]	Customize the Mipi instruction cache
cmdsLen	int	DL

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display.setCustomMipiCmds(
cmdsBuf, cmdsLen);
```

Remarks：
Restart is required after import

3.22 Set backlight parameters

Interface name:
int setBacklightParams(int screen, int Type, int TypeValue)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
Type	int	Type： ApiTool.BacklightType
TypeValue	int	Type: --Value descriptions for different Types Type_pwm_dc: -- 0: dc -- 1: pwm Type_inverse: -- 0: no inverse -- 1: inverse Type_max: --0-100, recommended>min Type_min: --0-100, recommended <max Type_pwm_freq: -- 100-100000 --Unit: Hz

Returned value：

Parameter name	Type	Explain
Returned value	int	0: Success -1: Failed

Code example:

```
int ret = SdkApi.getInstance().Display.setBacklightParams(
screen, Type, TypeValue);
```

Remarks：
-- None

3.23 Get backlight parameters

Interface name:
int getBacklightParams(int screen, int Type,ResultCallback<ResultInfo<Integer>> cb)

Parameter：

Parameter name	Type	Explain
screen	int	Display interface ID
Type	int	Type： ApiTool.BacklightType
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	int	syntony val： Type: --Value descriptions for different Types Type_pwm_dc: -- -- 0: dc -- 1: pwm Type_inverse: -- 0: no inverse -- 1: inverse Type_max: --0-100, recommended>min Type_min: --0-100, recommended <max Type_pwm_freq: -- 100-100000 --Unit: Hz

Code example:

```
SdkApi.getInstance().Display.getBacklightParams(
    screen, Type, new ResultCallback<ResultInfo<Integer>>() {
        @Override
        public void onMsg(Integer val) {

        }
});
```

Remarks：
-- None

3.24 Set screen display size

This interface is for bar screens/cutting screens

Interface name:
boolean setScreenGraphicDisplay(int scale, int x_pos, int y_pos, int width, int height)

Parameter：

Parameter name	Type	Explain
scale	int	Zoom switch: 0: Close 1: Open Default: On
x_pos	int	Relative horizontal offset to the top left
y_pos	int	Relative vertical offset to the top left corner
width	int	Display screen width
height	int	Display screen height

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret = SdkApi.getInstance().Display.
setScreenGraphicDisplay (scale, x, y, width, h eight);
```

Remarks：
Set and restart to take effect

3.25 Get screen display position

This interface is for bar screens/cutting screens

Interface name:
void getScreenGraphicDisplay(ResultCallback<ResultInfo<String>> cb)

Parameter：

Parameter name	Type	Explain
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	String	syntony val Format: scale, 0, 0, 1920,1080

Code example:

```
    SdkApi.getInstance().Display().getScreenGraphicDisplay (
        new ResultCallback<ResultInfo<String>>() {
            @Override
            public void onMsg(String val) {

            }
    })
```

Remarks：
-- None

4. System Control API

--Interface call method: SdkApi.getInstance(). SystemCtrl(). Interface

4.1 Restart the device

Interface name:
void deviceReboot()

Parameter：
-- None

Returned value：
-- None

Code example:

```
SdkApi.getInstance().SystemCtrl().deviceReboot()
```

Remarks：
-- None

4.2 Screenshot

Interface name:
void shotScreen(String path, int screen, ResultCallback<Boolean> cb)

Parameter：

Parameter name	Type	Explain
path	String	Screenshot file path and name
screen	int	Primary and secondary screens 0: Home screen 1: Secondary screen Default value: 0
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	syntony val： true： success false： fail

Code example:

```
SdkApi.getInstance().SystemCtrl().shotScreen("/mnt/sdcard/1.png",
0 , new ResultCallback<ResultInfo<Boolean>>() {
        @Override
        public void onMsg( Boolean val) {

        }
})
```

Remarks：
--The screenshot file format must be PNG. Currently, only main screen captures are supported.

4.3 Hide or show the navigation bar and status bar

Interface name:
void setStatusBarDisplay(boolean display)

Parameter：

Parameter name	Type	Explain
display	boolean	true: Navigation bar display false: Hide navigation bar

Returned value：
-- None

Code example:

```
SdkApi.getInstance().SystemCtrl().setStatusBarDisplay(false)
```

Remarks：
--Old interface, unified control of navigation bar and status bar
This interface is being phased out

4.4 Access the navigation bar and status bar

Interface name:
boolean getStatusBarDisplay()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true: Navigation bar display false: Hide navigation bar

Code example:

```
SdkApi.getInstance().SystemCtrl().getStatusBarDisplay()
```

Remarks：
--Old interface, unified navigation bar and status bar
This interface is being phased out

4.5 Hide or show the status bar

Interface name:
void setStatusBarDisplay2(boolean display)

Parameter：

Parameter name	Type	Explain
display	boolean	true: Status bar display false: hide status bar

Returned value：
-- None

Code example:

```
SdkApi.getInstance().SystemCtrl().setStatusBarDisplay2(false)
```

Remarks：
-- None

4.6 Get status bar status

Interface name:
boolean getStatusBarDisplay2()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true: Status bar display false: hide status bar

Code example:

```
boolean ret =
SdkApi.getInstance().SystemCtrl().getStatusBarDisplay2()
```

Remarks：
-- None

4.7 Hide or show the navigation bar

Interface name:
void setNavigationBar(boolean display)

Parameter：

Parameter name	Type	Explain
display	boolean	true: Navigation bar display false: Hide navigation bar

Returned value：
-- None

Code example:

```
SdkApi.getInstance().SystemCtrl().setNavigationBar(false)
```

Remarks：
-- None

4.8 Get navigation bar status

Interface name:
boolean getNavigationBar()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true: Navigation bar display false: Hide navigation bar

Code example:

```
boolean ret =
SdkApi.getInstance().SystemCtrl().getNavigationBar()
```

Remarks：
-- None

4.9 General standby mode

Interface name:
void setDeviceShallowOnoff(boolean onoff)

Parameter：

Parameter name	Type	Explain
onoff	boolean	true： starting up false： shut down

Returned value：
-- None

Code example:

```
SdkApi.getInstance().SystemCtrl().setDeviceShallowOnoff( true )
```

Remarks：
-- None

4.10 Get device standby status

Interface name:
boolean getDeviceShallowOnoff()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： starting up false： shut down

Code example:

```
boolean ret =
SdkApi.getInstance().SystemCtrl().getDeviceShallowOnoff()
```

Remarks：
-- None

4.11 Low-power standby mode

Interface name:
void setDeviceDeepStandby()

Parameter：
-- None

Returned value：
-- None

Code example:

```
SdkApi.getInstance().setDeviceDeepStandby()
```

Remarks：
--After executing this interface, the device will remain in standby mode until the next startup trigger condition is met.
--> Cut off power
--> Remote or button to turn on

4.12 Low-power timed standby for devices

Interface name:
void setDeviceDeepStandby(int time)

Parameter：

Parameter name	Type	Explain
time	int	Time to next startup Unit: minutes Range: 1-10080

Returned value：
-- None

Code example:

```
    //Turn on in 1 hour
    SdkApi.getInstance().SystemCtrl().setDeviceDeepStandby(1*60)
```

Remarks：
-- None

4.13 Set fan switch

Interface name:
boolean setFanOnoff(boolean onoff)

Parameter：

Parameter name	Type	Explain
onoff	boolean	true： open false： close Default: Off

Returned value：
-- None

Code example:

```
boolean ret = SdkApi.getInstance().SystemCtrl().setFanOnoff(true)
```

Remarks：
-- None

4.14 Get fan switch status

Interface name:
boolean getFanOnoff()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： open false： close

Code example:

```
boolean  ret = SdkApi.getInstance().SystemCtrl().getFanOnoff()
```

Remarks：
-- None

4.15 Silent App installation

Interface name:
void silentInstallApk(String path, boolean isAuto, ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
path	String	Absolute path for the app to install
isAuto	boolean	true: Start the app automatically after installation false: Install the app without launching it
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	syntony val： true： success false： fail

Code example:

```
SdkApi.getInstance().SystemCtrl().silentInstallApk("/mnt/sdcard/1
.apk", false, new ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

    }
})
```

Remarks：
Before using this interface, you need to consider the actual business scenario, because
When the installation file is stored on a USB drive, unplugging it after app installation may cause the app to crash. For this scenario, we recommend using the 4.16 interface.

4.16 Silent App Installation

Interface name:
void silentInstallApkForPackage(String path, String packageName, boolean isAuto, ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
path	String	Absolute path for the app to install
packageName	String	The app package name to install cannot be empty
isAuto	boolean	true: Start the app automatically after installation false: Install the app without launching it
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	syntony val： true： success false： fail

Code example:

```
SdkApi.getInstance().SystemCtrl().silentInstallApkForPackage("/mn
t/sdcard/1.apk", "com.xx.xx", false, new
ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

    }
});
```

Remarks：
-- None

4.17 Silent uninstallation of the app

Interface name:
void silentUninstallApk(String packageName, ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
packageName	String	App package name to uninstall
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	syntony val： true： success false： fail

Code example:

```
SdkApi.getInstance().SystemCtrl().silentUninstallApk(
"com.signway.testdemo",
new ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

    }
})
```

Remarks：
-- None

4.18 Get the last touch time

Interface name:
long getTPLastTouchTime()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	long	The last touch time, If the system does not receive a touch, the return value is 0.

Code example:

```
long time =
SdkApi.getInstance().SystemCtrl().getTPLastTouchTime()
```

Remarks：
-- None

4.19 Formatting Equipment

Interface name:
void formatStorageDevice(int Type, ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
Type	int	Format Type: --Local storage [0] --> ApiTool.Format.Internal --TF Card [1] --> ApiTool.Format.SDCard --USB drive [2] --> ApiTool.Format.Usb
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	syntony val： true： success false： fail

Code example:

```
Format local storage
SdkApi.getInstance().SystemCtrl().getTPLastTouchTime( 0 ,
new ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

    }
})
```

Remarks：
--The SDK version must be 28 or later. Formatting local storage is supported only.

4.20 Adjust engine startup animation volume

Interface name:
long setBootVideoVolume(int volume)

Parameter：

Parameter name	Type	Explain
volume	int	Volume level Range: 0-100

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
    Set the engine animation video volume to 50%
    Boolean ret =
    SdkApi.getInstance().SystemCtrl().setBootVideoVolume(50)
```

Remarks：
-- None

4.21 Restore factory settings

Interface name:
int recoverySystem(int Type)

Parameter：

Parameter name	Type	Explain
Type	int	0: Restore all system parameters, including locally stored files

Returned value：

Parameter name	Type	Explain
Returned value	int	0: Success -1: Failed

Code example:

```
int ret =
SdkApi.getInstance().SystemCtrl().recoverySystem(0)
```

Remarks：
-- None

4.22 Get Total Boot Duration

Interface name:
long getUpTime()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	`long`	Total `Duration`

Code example:

```
long uptime = SdkApi.getInstance().SystemCtrl().getUpTime();
```

Remarks：
-- None

4.23 Reset Total Boot Duration

Interface name:
void resetUpTime(long msec)

Parameter：

Parameter name	Type	Explain
Returned value	`long`	Total Duration

Returned value：
-- None

Code example:

```
SdkApi.getInstance().SystemCtrl().resetUpTime(mSec);
```

Remarks：
-- None

4.24 Get NTP Server Address

Interface name:
String getNtpServerAddress()

Parameter :
-- None

Returned value:

Parameter name	Type	Explain
Returned value	`String`	`NTP address string`

Code example:

```
String address =
SdkApi.getInstance().SystemCtrl().getNtpServeSdress();
```

Remarks：
-- None

4.25 Reset NTP Server Address

Interface name:

`void setNtpServerAddress(String address)`

Parameter：

Parameter name	Type	Explain
Returned value	`String`	`New NTP address`

Returned value:
-- None

Code example:

```
SdkApi.getInstance().SystemCtrl().setNtpServerAddress(address);
```

Remarks：
-- None

5. Timer API

--Interface call method: SdkApi.getInstance(). TimerSwitch(). Interface

--Function description: Turn on during the time period and turn off outside the time period

Instance：

Turn on at 9:00, turn off at 10:00, and the device operates from 9:00 to 10:00

The device starts at 10:00 and shuts down at 9:00. The operating time is from 10:00 to 9:00 the next day.

The boot time equals the shutdown time. This invalid setting maintains the current status.

5.1 Set timer switch

Interface name:
void setTimerSwitchOnoff(boolean onoff)

Parameter：

Parameter name	Type	Explain
onoff	boolean	Timer function switch Default: Off

Returned value：
-- None

Code example:

```
SdkApi.getInstance().TimerSwitch().setTimerSwitchOnoff(true)
```

Remarks：
The timer starts 2 minutes after the device is powered on or the system restarts. The timer does not take effect within the first 2 minutes of power-on.

5.2 Get the timer switch status

Interface name:
boolean getTimerSwitchOnoff()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： open false： close

Code example:

```
boolean ret =
SdkApi.getInstance().TimerSwitch().getTimerSwitchOnoff()
```

Remarks：
-- None

5.3 Set a scheduled startup time

Interface name:
boolean setTimerOnTime(String onTime)

Parameter：

Parameter name	Type	Explain
onTime	String	Scheduled on-time Format: "HH: mm"

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret =
SdkApi.getInstance().TimerSwitch().setTimerOnTime("12:01")
```

Remarks：
-- None

5.4 Clear scheduled startup time

Interface name:
void clearTimerOnTime()

Parameter：
-- None

Returned value：
-- None

Code example:

```
SdkApi.getInstance().TimerSwitch().clearTimerOnTime()
```

Remarks：
--Default time: 00:00

5.5 Set scheduled startup time

Interface name:
String getTimerOnTime()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	Scheduled on-time Format: "HH: mm"

Code example:

```
String time =
SdkApi.getInstance().TimerSwitch().getTimerOnTime()
```

Remarks：
--Default time: 00:00

5.6 Set scheduled shutdown time

Interface name:
boolean setTimerOffTime(String offTime)

Parameter：

Parameter name	Type	Explain
offTime	String	Scheduled shutdown time Format: "HH: mm"

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
boolean ret =
SdkApi.getInstance().TimerSwitch().setTimerOffTime("12:01")
```

Remarks：
-- None

5.7 Clear scheduled shutdown time

Interface name:
void clearTimerOffTime()

Parameter：
-- None

Returned value：
-- None

Code example:

```
SdkApi.getInstance().TimerSwitch().clearTimerOffTime()
```

Remarks：
--Default time: 00:00

5.8 Set scheduled shutdown time

Interface name:
String getTimerOffTime()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	Scheduled shutdown time Format: "HH: mm"

Code example:

```
String time =
SdkApi.getInstance().TimerSwitch().getTimerOffTime()
```

Remarks：
--Default time: 00:00

5.9 Set timer shutdown mode

Interface name:
void setTimerStandbyMode(boolean standby)

Parameter：

Parameter name	Type	Explain
standby	boolean	false: Normal standby true: Low-power standby Default value: false

Returned value：
-- None

Code example:

```
SdkApi.getInstance().TimerSwitch().setTimerStandbyMode(false)
```

Remarks：
-- None

5.10 Get scheduled shutdown mode

Interface name:
boolean getTimerStandbyMode()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	false: Normal standby true: Low-power standby

Code example:

```
boolean ret =
SdkApi.getInstance().TimerSwitch().getTimerOffTime()
```

Remarks：
-- None

6. Dog-related API

--Interface call method: SdkApi.getInstance(). Watchdog(). Interface

Function: Watchdog reset. By default, the SDK performs watchdog feeding. When the App performs watchdog feeding, the SDK initiates the App watchdog feeding service and monitors the App watchdog feeding.

6.1 Set the dog feeder

Interface name:
boolean setWatchdogOnoff(boolean onoff)

Parameter：

Parameter name	Type	Explain
onoff	boolean	Is the dog feeder an app or a system true：APP false： system Default value: false

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
SdkApi.getInstance().Watchdog().setWatchdogOnoff(true)
```

Remarks：
-- None

6.2 Get dog feeder

Interface name:
boolean getWatchdogOnoff()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true：APP false： system

Code example:

```
SdkApi.getInstance().Watchdog ().get WatchdogOnoff ()
```

Remarks：
-- None

6.3 Set the watchdog timeout

Interface name:
boolean setWatchdogTimeout(int timeout)

Parameter：

Parameter name	Type	Explain
timeout	int	App watchdog timeout Default: 3 minutes Duration: 30 seconds to 5 minutes Unit: seconds

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
    Set the dog feeding timeout to 60 seconds

    SdkApi.getInstance().Watchdog().setWatchdogTimeout(60)
```

Remarks：
-- None

6.4 Feed the Dog

Interface name:
boolean feedWatchdog()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
SdkApi.getInstance().Watchdog ().feedWatchdog ()
```

Remarks：
-- None

7. System time-related API

--Interface call method: SdkApi.getInstance(). SystemTime(). Interface

7.1 Set system time

Interface name:
void setSystemTime(long time)

Parameter：

Parameter name	Type	Explain
time	long	Time difference from 1970 Unit: milliseconds

Returned value：
-- None

Code example:

```
SdkApi.getInstance().SystemTime().setSystemTime(60*1000*1000)
```

Remarks：
-- None

7.2 Get RTC Time

Interface name:
String getRtcTime()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	RTC time Time format: "yyyy/MM/dd hh:mm:ss"

Code example:

```
String time =
SdkApi.getInstance().TimerSwitch().getRtcTime ()

Return value example
2025/01/01 12:00:00
```

Remarks：
-- None

7.3 Set automatic calibration time

Interface name:
void setAutoTimeOnoff(boolean onoff)

Parameter：

Parameter name	Type	Explain
onoff	boolean	true: NTP enabled false: NTP is disabled

Returned value：
-- None

Code example:

```
SdkApi.getInstance().SystemTime().setAutoTimeOnoff(true)
```

Remarks：
-- None

7.4 Get the auto-calibration time switch status

Interface name:
boolean getAutoTimeOnoff()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true: NTP enabled false: NTP is disabled

Code example:

```
SdkApi.getInstance().SystemTime().getAutoTimeOnoff()
```

Remarks：
-- None

8. Hardware-related API

--Interface call method: SdkApi.getInstance(). Hardware(). Interface

8.1 Get GPIO input level

Interface name:
int getGpioInput(int group, int num)

Parameter：

Parameter name	Type	Explain
group	int	The group of the GPIO; For example, the group value of GPIO0_B3 is GPIO0. See "GPIO Value Definition"
num	int	The group ID of the GPIO; For example, GPIO0_B3, where the num value is B3 See "GPIO Value Definition"

Returned value：

Parameter name	Type	Explain
Returned value	boolean	1: High level 0: Low level -1: Read error

Code example:

```
//GPIO0_B3
int level = SdkApi.getInstance().Hardware().getGpioInput(
ApiTool.GpioPin.GPIO0, ApiTool.GpioPin.PB3);
```

Remarks：
-- None

8.2 Configure GPIO Output Status

Interface name:
boolean setGpioOutput(int group, int num, boolean level)

Parameter：

Parameter name	Type	Explain
group	int	The group of the GPIO; For example, the group value of GPIO0_B3 is GPIO0. See "GPIO Value Definition"
num	int	The group ID of the GPIO; For example, GPIO0_B3, where the num value is B3 See "GPIO Value Definition"
level	boolean	true： high level false: low level

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
//GPIO0_B3
int level = SdkApi.getInstance().Hardware().setGpioOutput(
ApiTool.GpioPin.GPIO0, ApiTool.GpioPin.PB3, true);
```

Remarks：
-- None

8.3 Read flash data

Interface name:
void flashRead(int Type, int len, ResultCallback<ResultInfo<Integer>> cb)

Parameter：

Parameter name	Type	Explain
Type	int	data Type -- HWID --Authorization information, etc. see details ApiTool.FlashRWType Category
len	int	Length of data to read
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	String	syntony val1: Read data val2: The actual length of the read

Code example:

```
SdkApi.getInstance().Hardware().flashRead(
ApiTool.FlashRWType.HWID, 1,
new ResultCallback<ResultInfo<Interger>>() {
    @Override
    public void onMsg(ResultInfo<Integer> val) {
        byte[] array = val.getArrayByte();
        int len = val.getVal();
    }
});
```

Remarks：
Read the entire data block of the Type
--Maximum data length: 1024 bytes; some Types are only a few bytes

8.4 Write Flash Data

Interface name:
void flashWrite(int Type, int len, int[] buffer, ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
Type	int	data Type -- HWID --Authorization information, etc. see details ApiTool.FlashRWType Category
len	int	Write data length
buffer	int[]	Write to data buffer
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	true： success false： fail

Code example:

```
SdkApi.getInstance().Hardware().flashWrite(
ApiTool.FlashRWType.APP_INFO, 1, buffer
new ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

    }
});
```

Remarks：
-- None

8.5 Mainboard operation indicator light

Interface name:
void setRunningLedOnoff(boolean onoff, ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
onoff	boolean	true： open false： close
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	Get result callback: true： success false： fail

Code example:

```
    Turn on the running indicator
    SdkApi.getInstance().Hardware().setRunningLedOnoff(true,
    new ResultCallback<ResultInfo<Boolean>>() {
        @Override
        public void onMsg(Boolean val) {

        }
    });
```

Remarks：
--Different motherboard configurations have differences, and there are
--> The motherboard has no indicator lights
--> The indicator light is determined by the hardware circuit
--Default: On

8.6 Access motherboard status indicator

Interface name:
void getRunningLedOnoff(ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	String	Get result callback: true： open false： close

Code example:

```
SdkApi.getInstance().Hardware().getRunningLedOnoff(
new ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

    }
});
```

Remarks：
--Different motherboard configurations have differences, and there are
--> The motherboard has no indicator lights
--> The indicator light is determined by the hardware circuit
--Default: On

8.7 Set the red indicator light switch on the keypad

Interface name:
void setKeyBoardLedROnoff(boolean onoff, ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
onoff	boolean	true： open false： close
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	Get result callback: true： success false： fail

Code example:

```
SdkApi.getInstance().Hardware().setKeyBoardLedROnoff(true,
new ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

    }
});
```

Remarks：
--Default control follows the LED common cathode design based on the main board

8.8 Access the red indicator switch on the keypad

Interface name:
void getKeyBoardLedROnoff(ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	Retrieve status true： open false： close

Code example:

```
SdkApi.getInstance().Hardware().getKeyBoardLedROnoff(
new ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

    }
});
```

Remarks：
--Default control follows the LED common cathode design based on the main board

8.9 Set the green indicator light switch on the keypad

Interface name:
void setKeyBoardLedGOnoff(boolean onoff,ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
onoff	boolean	true： open false： close
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	Get result callback: true： success false： fail

Code example:

```
SdkApi.getInstance().Hardware().setKeyBoardLedGOnoff(true,
new ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

        }
});
```

Remarks：
--Default control follows the LED common cathode design based on the main board

8.10 Access the green indicator switch on the keypad

Interface name:
void getKeyBoardLedGOnoff(ResultCallback<ResultInfo<Boolean>> cb)

Parameter：

Parameter name	Type	Explain
cb	class	ResultCallback class

Returned value：

Parameter name	Type	Explain
Return parameter val	boolean	Retrieve status true： open false： close

Code example:

```
SdkApi.getInstance().Hardware().getKeyBoardLedGOnoff(
new ResultCallback<ResultInfo<Boolean>>() {
    @Override
    public void onMsg(Boolean val) {

    }
});
```

Remarks：
--Default control follows the common cathode design for LED, based on the main board

8.11 Set the USB port power switch

Interface name:
boolean setUsbPortPowerOnoff(int usbPort, boolean onoff)

Parameter：

Parameter name	Type	Explain
usbPort	int	USB port index number
onoff	boolean	true： open false： close

Returned value：
-- None
Code example:

```
SdkApi.getInstance().Hardware().setUsbPortPowerOnoff(port, true);
```

Remarks：
--No such interface
The USB port index differs based on the motherboard.

8.12 Access USB port power switch

Interface name:
boolean getUsbPortPowerOnoff(int usbPort)

Parameter：

Parameter name	Type	Explain
usbPort	int	USB port index number

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： open false： close

Code example:

```
SdkApi.getInstance().Hardware().getUsbPortPowerOnoff(port);
```

Remarks：
--No such interface
The USB port index differs based on the motherboard.

8.13 Set speaker power parameters

Interface name:
boolean setSpeakerConfigsSave(boolean status, int power, int resistance)

Parameter：

Parameter name	Type	Explain
status	boolean	Loudspeaker parameter status true： enable false： close Default value: the default configuration of the board
power	int	Speaker power: 1-10W
resistance	int	Loudspeaker resistance: 2, 4, 6, 8Ω

Returned value：

Parameter name	Type	Explain
Returned value	boolean	true： success false： fail

Code example:

```
Set the speaker to 2W 4Ω
boolean ret =
SdkApi.getInstance().Hardware().setSpeakerConfigsSave(status,
2 ,  4);
```

Remarks：
-- None

8.14 Get speaker power parameters

Interface name:
void getSpeakerConfigs(ResultCallback<ResultInfo<Integer>> cb)

Parameter：

Parameter name	Type	Explain
cb	class	Get speaker power back

Returned value：

Parameter name	Type	Explain
Return parameter val	int	Retrieve array with callback: The first data is status The second data is the speaker power The third data is the speaker resistance

Code example:

```
SdkApi.getInstance().Hardware().getSpeakerConfigs(
new ResultCallback<ResultInfo<Integer>>() {
    ...
    byte[] array = val.getArrayByte();
    array[0] //status
    array[1] //power
    array[2] //Resistance
});
```

Remarks：
-- None

8.15 Open serial port device 1

--Only supports opening the serial port in 8N1 mode

Interface name:
int openSerial(String serialName, int baud)

Parameter：

Parameter name	Type	Explain
serialName	String	Serial port device name Example: "/dev/ttyS0"
baud	int	Serial port baud rate

Returned value：

Parameter name	Type	Explain
Returned value	int	Serial port device handle

Code example:

```
int serial Fd =
SdkApi.getInstance().Hardware().openSerial("/dev/ttyS0", 115200);
```

Remarks：
-- None

8.16 Open Serial Port 2

-- universal method

Interface name:
int openSerial(String serialName, int baud, int dataBits, int parity, int stopBits)

Parameter：

Parameter name	Type	Explain
serialName	String	Serial port device name Example: "/dev/ttyS0"
baud	int	Serial port baud rate
dataBits	int	data bit Values: 5, 6, 7, 8
parity	int	even-odd check 0：NONE 1：ODD 2：EVEN
stopBits	int	stop bit Value: 1, 2

Returned value：

Parameter name	Type	Explain
Returned value	int	Serial port device handle

Code example:

```
int serialFd =
SdkApi.getInstance().Hardware().openSerial("/dev/ttyS0", 115200,
8, 0, 1);
```

Remarks：
-- None

8.17 Close serial port

Interface name:
int closeSerial(int serial)

Parameter：

Parameter name	Type	Explain
serial	int	Serial port device handle

Returned value：

Parameter name	Type	Explain
Returned value	int	0: Closed successfully Other values: Failed to disable

Code example:

```
int ret = SdkApi.getInstance().Hardware().closeSerial(serial);
```

Remarks：
-- None

8.18 Read serial port data

Interface name:
int readSerial(int serial, byte[] buf, int len)

Parameter：

Parameter name	Type	Explain
serial	int	Handle for the serial port
buf	byte[]	Save the cached data
len	int	Length of data to read

Returned value：

Parameter name	Type	Explain
Returned value	int	Actual data read length

Code example:

```
int readLen = SdkApi.getInstance().Hardware().readSerial(serial,
buf, len);
```

Remarks：
-- None

8.19 Write serial port data

Interface name:
int writeSerial(int serial, byte[] buf, int len)

Parameter：

Parameter name	Type	Explain
serial	int	Handle for the serial port
buf	byte[]	Write data cache
len	int	Length of data to write

Returned value：

Parameter name	Type	Explain
Returned value	int	Actual data written length

Code example:

```
int writeLen =
SdkApi.getInstance().Hardware().writeSerial(serial, buf, len);
```

Remarks：
-- None

8.20 Configure and enable the CAN bus

Interface name:
int canBusConfigUp(int bitrate)

Parameter：

Parameter name	Type	Explain
bitrate	int	CAN bus rate

Returned value：

Parameter name	Type	Explain
Returned value	int	0: Success Other values: Failed

Code example:

```
int ret = SdkApi.getInstance().Hardware().canBusConfigUp(250000);
```

Remarks：
-- None

8.21 Open CAN device

Interface name:
int canOpen(int canId)

Parameter：

Parameter name	Type	Explain
canId	int	can highway id

Returned value：

Parameter name	Type	Explain
Returned value	int	CAN device handle

Code example:

```
int ret = SdkApi.getInstance().Hardware().canOpen(0);
```

Remarks：
--The ID defaults to 0

8.22 Send CAN frame data

Interface name:
int canSend(int can, int frameFlag, int frameId, byte[] canData, int len)

Parameter：

Parameter name	Type	Explain
can	int	CAN device handle
frameFlag	int	frame Type For example: CAN_EFF_FLAG
frameId	int	Frame identifier
canData	byte[]	Frame Data
len	int	Frame data length Maximum 8 bytes

Returned value：

Parameter name	Type	Explain
Returned value	int	CAN device handle

Code example:

```
int ret =
SdkApi.getInstance().Hardware().canSend(can,0,0x123,0x11,1);
```

Remarks：
-- None

8.23 Send CAN frame data

Interface name:
int canRecv(int can, int[] canInfo)

Parameter：

Parameter name	Type	Explain
can	int	CAN device handle
canInfo	int[]	Retrieved data canInfo[0] is the can_id canInfo[1] is can_dlc canInfo[2] is the starting data of can_data

Returned value：

Parameter name	Type	Explain
Returned value	int	The actual data length of the entire canInfo

Code example:

```
    int[] canInfo = new int[16];
    int ret = SdkApi.getInstance().Hardware().canRecv(can, canInfo);
```

Remarks：
-- None

8.24 Disable CAN device

Interface name:
int canClose(int can)

Parameter：

Parameter name	Type	Explain
can	int	CAN device handle

Returned value：

Parameter name	Type	Explain
Returned value	int	0: Success Other values: Failed

Code example:

```
int ret = SdkApi.getInstance().Hardware().canClose(can);
```

Remarks：
-- None

8.25 Get remote control status

Interface name:
int getRcuState()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	`int`	0: Open 1: Close all 2: All except for the power button are turned off

Code example:

```
int ret = SdkApi.getInstance().Hardware().getRcuState();
```

Remarks：
-- None

8.26 Set remote control status

Interface name:
void setRcuState(int state)

Parameter:

Parameter name	Type	Explain
can	int	Remote control status value 0: Open 1: Close all 2: All except for the power button are turned off

Returned value：
-- None

Code example:

```
int rcuState = 0;
SdkApi.getInstance().Hardware().setRcuState(rcuState);
```

Remarks：
-- None

8.27 Obtain the serial port power switch

Interface name:
int getSerialPortOnOff(String serialName)

Parameter:

Parameter name	Type	Explain
serialName	String	Serial device name Example: "/dev/ttyS0"

Returned value:

Parameter name	Type	Explain
Returned value	int	

Code example:

```
int ret =
SdkApi.getInstance().Hardware().getSerialPortOnOff("/dev/ttyS0");
```

Remarks：
-- None

8.28 Set serial port power switch

Interface name:
boolean setSerialPortOnOff(String serialName, Boolean onOff);

Parameter:

Parameter name	Type	Explain
serialName	String	Serial device name Example: "/dev/ttyS0"
onOff	boolean	True: Open False: Close

Returned value:

Parameter name	Type	Explain
Returned value	boolean	True: Successful False: Failed

Code example:

```
boolean ret =
SdkApi.getInstance().Hardware().setSerialPortOnOff("/dev/ttyS0",
true);
```

Remarks：
-- None

9. Other related APIs

--Interface call method: SdkApi.getInstance(). Other(). Interface

9.1 Obtain the ICCID number of the SIM card

Interface name:
String getSimICCID()

Parameter：
-- -- None

Returned value：

Parameter name	Type	Explain
Returned value	String	val: iccid

Code example:

```
String iccid = SdkApi.getInstance().Other().getSimICCID();
```

Remarks：
The SIM card must be properly inserted into the 4G module.

9.2 Get IMEI

Interface name:
String getModemIMEI()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	val: imei

Code example:

```
String imei = SdkApi.getInstance().Other().getModemIMEI();
```

Remarks：
The SIM card must be properly inserted into the 4G module.

9.3 Execute shell commands

Interface name:
void execShellCommand(String cmd)

Parameter：

Parameter name	Type	Explain
cmd	String	shell order Example: input keyevent 26

Returned value：
-- None

Code example:

```
int ret =
SdkApi.getInstance().Other ().execShellCommand ("input keyevent
26");
```

Remarks：
-- None

9.4 Read emmc lifespan

Interface name:
String getEmmcLeftTime()

Parameter：
-- None

Returned value：

Parameter name	Type	Explain
Returned value	String	val: EMMC remaining life Example: 010101

Code example:

```
String leftTime = SdkApi.getInstance().Other().getEmmcLeftTime();
```

Remarks：
-- None

10. Scale API

--Based on the actual motherboard, some motherboards do not have scale chips and do not support this Type of interface.

10.1 Register the scale listener

Interface name:
int registerScaleListener(int appId, ScaleListener listener)

Parameter：

Parameter name	Type	Explain
appId	int	app Type Id Define in ApiTool.ScaleAppId
listener	ScaleListener	Listening listener

Returned value：

Parameter name	Type	Explain
Returned value	String	Registration successful ApiTool.API_RES_*

Code example:

```
int registerScaleListener(ApiTool.ScaleAppId.ID_APP_CUSTOM, new
ScaleListener() { });
```

Remarks：
-- None

10.2 Unregister the Scale Listener

Interface name:
int unregisterScaleListener(int appId, ScaleListener listener)

Parameter：

Parameter name	Type	Explain
appId	int	app Type ID Define in ApiTool.ScaleAppId
listener	ScaleListener	Listening listener

Returned value：

Parameter name	Type	Explain
Returned value	String	Is the cancellation successful? ApiTool.API_RES_*

Code example:

```
int unregisterScaleListener(ApiTool.ScaleAppId.ID_APP_CUSTOM, new
ScaleListener() { });
```

Remarks：
-- None

10.3 Set scale to execute events

Interface name:
int setScaleEvent(int appId, int actionId, String strParam, int[] dataArray, int arrayLen)

Parameter：

Parameter name	Type	Explain
appId	int	app Type Id Define in ApiTool.ScaleAppId
actionId	int	event Id Define in ApiTool.ScaleEvent
strParam	String	The parameter can be null
dataArray	int[]	The parameter can be null If null, arrayLen must be 0
arrayLen	int	Data array length

Returned value：

Parameter name	Type	Explain
Returned value	String	Set successfully Event results are notified to the listener

Code example:

```
int setScaleEvent(ApiTool.ScaleAppId.ID_APP_CUSTOM,
ApiTool.ScaleEvent.QUERY_VERSION);
```

Remarks：
-- None

10.4 actionId command set

command name	command code	Explain	remarks
SYSTEM_EVENT_GET_SCALE_VERSION	0x8001	Get the scaler version	Set parameters 【dataArray[0]】： -- **int**[] dataArray = {1}; Return value【dataArray】： -- Example: **new** String(dataArray, 0, dataLen)
SYSTEM_EVENT_SET_SYSTEM_SHUTDOWN	0x8002	safe shutdown	Set parameters 【dataArray[0]】： -- **int**[] dataArray = {1}; Return value 【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
~~***SYSTEM_EVENT_SET_SYSTEM_POWER_ON OFF***~~	~~0x8003~~	Host restart	
SYSTEM_EVENT_SET_SYSTEM_REBOOT	0x8004	Get the scaler version	Set parameters 【dataArray[0]】： -- **int**[] dataArray = {1}; Return value 【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
SYSTEM_EVENT_GET_SCALE_TYPE	0x8005	Obtain product definition	Set parameters 【dataArray[0]】： -- **int**[] dataArray = {1}; Return value 【dataArray[0]】： 0x01：L 0x02：V 0x03：L2 0x04：VL 0x05：HL 0x06：S 0x07：SL 0x08：D 0x09：DL

			0x0a：F 0x0b：FL 0x0c：HV 0x0d：HL1
		Get screen parameters	
~~SCALE_EVENT_GET_PANEL_PARAM~~	~~0x3001~~	~~Set screen parameters~~	Set parameters【dataArray】： ~~int[] dataArray = {1};~~ ~~Return value【dataArray】：~~ ~~--0x01: Represents screen type 1~~ ~~--0x02: Represents screen type 1~~
~~SCALE_EVENT_SET_PANEL_PARAM~~	~~0x3002~~	~~Get screen switch status~~	
SCALE_EVENT_GET_SCREEN_ONOFF	0x3003	Obtain product definition	Set parameters 【dataArray[0]】： -- int[] dataArray = {1}; Return value 【dataArray[0]】： --0x01: Screen status is on --0x00: Screen status is off
SCALE_EVENT_SET_SCREEN_ONOFF	0x3004	Set screen switch status	Set parameters 【dataArray[0]】： --0x01: Screen status is on --0x00: Screen status is off Return value 【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
SCALE_EVENT_GET_INPUT_SOURCE	0x3005	Obtain signal source	Set parameters 【dataArray[0]】： -- **int**[] dataArray = {1}; Return value 【dataArray[0]】： -- 0x01：Android -- 0x02：Hdmi
SCALE_EVENT_SET_INPUT_SOURCE	0x3006	Set signal source	Set parameters 【dataArray[0]】： -- 0x01：Android -- 0x02：Hdmi Return value 【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
SCALE_EVENT_GET_BACKLIGHT_ONOFF	0x3007	Get backlight switch	Set parameters 【dataArray[0]】： -- **int**[] dataArray = {1};

			Return value 【dataArray[0]】: -- 0x01: Backlight status is on -- 0x00: Backlight status is off
SCALE_EVENT_SET_BACKLIGHT_ONOFF	0x3008	Set backlight switch	Set parameters 【dataArray[0]】: -- 0x01: Backlight status is on -- 0x00: Backlight status is off Return value 【dataArray[0]】: -- 0x10: Success -- 0x11: Failed
SCALE_EVENT_GET_OSD_ORIENTATION	0x3009	Get OSD direction	Set parameters【dataArray】: -- **int**[] dataArray = {1}; Return value 【dataArray[0]】: -- 0x01: 0° -- 0x02: 90° -- 0x03: 180° -- 0x04: 270°
SCALE_EVENT_SET_OSD_ORIENTATION	0x300A	Set OSD direction	Set parameters 【dataArray[0]】: -- 0x01: 0° -- 0x02: 90° -- 0x03: 180° -- 0x04: 270° Return value 【dataArray[0]】: -- 0x10: Success -- 0x11: Failed
SCALE_EVENT_REPORT_INPUT_SOURCE	0x300B	Report signal channel	Give callbacks to all registered applications Return value 【dataArray[0]】: -- 0x01: Android -- 0x02: Hdmi
~~***SYSTEM_EVENT_SET_FUNCTION_KEYS***~~	~~0x300C~~	~~Send function key~~	
SCALE_EVENT_REPORT_SCREEN_TYPE	0x300D	Report product type	Give callbacks to all registered applications Return value 【dataArray[0]】: 0x01: L 0x02: V 0x03: L2 0x04: VL 0x05: HL 0x06: S

			0x07：SL 0x08：D 0x09：DL 0x0a：F 0x0b：FL 0x0c：HV 0x0d：HL1
SCALE_EVENT_REPORT_SCREEN_VOLUME	0x300E	Report volume	Give callbacks to all registered applicationsReturn value【dataArray[0]】
~~***SCALE_EVENT_GET_BACKLIGHT***~~	~~0x3010~~	~~Obtain backlight size~~	
~~***SCALE_EVENT_SET_BACKLIGHT***~~	~~0x3011~~	~~Set backlight size~~	
~~***SCALE_EVENT_QUERY_MUTE_STATUS***~~	~~0x3012~~	~~Get MUTE (mute) status~~	
~~***SCALE_EVENT_SET_MUTE_STATUS***~~	~~0x3013~~	~~(Hardware MUTE)~~	
~~***SCALE_EVENT_QUERY_OSD_LANGUAGE***~~	~~0x3014~~	~~Set MUTE (mute) status~~	
~~***SCALE_EVENT_SET_OSD_LANGUAGE***~~	~~0x3015~~	~~(Hardware MUTE)~~	
AUDIO_EVENT_GET_VOLUME	0x3101	Get volume level	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value【dataArray[0]】
AUDIO_EVENT_SET_VOLUME	0x3102	Set the volume level	Set parameters【dataArray】： -- 0 - 100 Return value【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
AUDIO_EVENT_GET_MUTE_STATUS	0x3103	Get MUTE (mute) status	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value【dataArray[0]】： --0x00: Represents the mute state is off (not muted) --0x01: Represents the mute state is on (muted)
AUDIO_EVENT_SET_MUTE_STATUS	0x3104	(Software MUTE)	Set parameters【dataArray】： --0x00: Represents the mute state is off (not muted) --0x01: Represents the mute state is on (muted) Return value【dataArray[0]】： -- 0x10：Success

			-- 0x11：Failed
AUDIO_EVENT_GET_HORN_OUTPUT_POWER	0x3107	Set MUTE (mute) status	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value【dataArray】： --DataArray [0]: represents the resistance value of the speaker in ohms, usually 4/8 --DataArray [1]: Speaker power wattage, generally ranging from 1 to 10
AUDIO_EVENT_SET_HORN_OUTPUT_POWER	0x3108	(Software MUTE)	Set parameters【dataArray】： --DataArray [0]: represents the resistance value of the speaker in ohms, usually 4/8 --DataArray [1]: Speaker power wattage, generally ranging from 1 to 10 Return value【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
PICTURE_EVENT_GET_BACKLIGHT_LEVEL	0x3201	Obtain the output power of the speaker	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value【dataArray[0]】：
PICTURE_EVENT_SET_BACKLIGHT_LEVEL	0x3202	Set the output power of the speaker	Set parameters【dataArray】： Return value【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
PICTURE_EVENT_GET_CONTRAST	0x3203	Obtain image brightness	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value【dataArray[0]】： -- 0 - 100
PICTURE_EVENT_SET_CONTRAST	0x3204	Set image brightness	
PICTURE_EVENT_GET_SHARPNESS	0x3205	Obtain contrast	
PICTURE_EVENT_SET_SHARPNESS	0x3206	Set contrast	
PICTURE_EVENT_GET_COLOR_TEMP	0x3207	empty	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value【dataArray[0]】： -- 0x01：warm

			-- 0x02：standard -- 0x03：cool -- 0x04：user
PICTURE_EVENT_SET_COLOR_TEMP	0x3208	empty	Set parameters【dataArray】： -- 0x01：warm -- 0x02：standard -- 0x03：cool -- 0x04：user Return value 【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
PICTURE_EVENT_GET_R_GAIN	0x3209	Obtain color temperature	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value 【dataArray[0]】： -- 0 - 255
PICTURE_EVENT_SET_R_GAIN	0x320A	Set color temperature	Set parameters【dataArray】： -- 0 - 255 Return value 【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
PICTURE_EVENT_GET_G_GAIN	0x320B	Obtain R Gain	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value 【dataArray[0]】： -- 0 - 255
PICTURE_EVENT_SET_G_GAIN	0x320C	Set R Gain	Set parameters【dataArray】： -- 0 - 255 Return value 【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
PICTURE_EVENT_GET_B_GAIN	0x320D	Get G Gain	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value 【dataArray[0]】： -- 0 - 255
PICTURE_EVENT_SET_B_GAIN	0x320E	Set G Gain	Set parameters【dataArray】： -- 0 - 255 Return value 【dataArray[0]】： -- 0x10：Success

			-- 0x11：Failed
		Obtain B Gain	
SCALE_EVENT_SET_LOGO_STYLE	0x3304	Set B Gain	Set parameters【dataArray】： -- 0x00：Android -- 0x02：Goodview -- 0x03：Videri Return value 【dataArray[0]】： -- 0x10：Success -- 0x11：Failed
SCALE_EVENT_GET_LOGO_STYLE	0x3305	Set logo style	Set parameters【dataArray】： -- int[] dataArray = {1}; Return value 【dataArray[0]】： -- 0x00：Android -- 0x02：Goodview -- 0x03：Videri

10.4.1 Report Screen Reference Index

Run code example

```
int setScaleEvent(ApiTool.ScaleAppId.ID_APP_CUSTOM,
ApiTool.ScaleEvent. SCALE_EVENT_REPORT_SCREEN_Type);
```

Error Handling Code Example

```
ScaleListener scaleListener = new ScaleListener() {
    @Override
    public void onNotify(int actionId, byte[] dataArray, int
dataLen) {
          switch (actionId) {
              case SCALE_EVENT_REPORT_SCREEN_Type:
                   //todo your things
                   break;
         }
    }
};

Method description:
Rewrite the ScaleListener callback interface and call
registerScaleListener
Return parameter description:
-- actionId : SCALE_EVENT_REPORT_SCREEN_Type
--dataArray: Returns an array of data [length 1 byte], with
specific values to be defined
```

10.4.2 Report screen brightness

Run code example

```
int setScaleEvent(ApiTool.ScaleAppId.ID_APP_CUSTOM,
ApiTool.ScaleEvent.SCALE_EVENT_REPORT_SCREEN_VOLUME);
```

Error handling code example

```
ScaleListener scaleListener = new ScaleListener() {
    @Override
    public void onNotify(int actionId, byte[] dataArray, int
dataLen) {
          switch (actionId) {
              case SCALE_EVENT_REPORT_SCREEN_VOLUME:
                   //todo your things
                   break;
          }
    }
};

Method description:
Rewrite the ScaleListener callback interface and call
registerScaleListener
Return parameter description:
-- actionId : SCALE_EVENT_REPORT_SCREEN_VOLUME
--dataArray: Returns an array of data [length 1 byte], with values
ranging from 0 to 100
```

11. CommonEvent-related API

11.1 Register a Common listener

Interface name:
int registerCommonListener(int appId, ISdkCommonListener listener)

Parameter：

Parameter name	Type	Explain
appId	int	app Type Id Defined in ApiTool.CommonAppId
listener	ScaleListener	Listening listener

Returned value：

Parameter name	Type	Explain
Returned value	String	Registration successful ApiTool.API_RES_*

Code example:

```
int registerCommonListener(ApiTool.CommonAppId.ID_APP_CUSTOM, new
ISdkCommonListener() { });

private ISdkCommonListener commonListener = new
ISdkCommonListener.Stub() {
        @Override
        public void CommonListenerNotify(int actionId, String
sVal, int[] iArray, int intArrayLen) throws RemoteException {
            if (actionId == ApiTool.CommonEvent.SWITCH_ETH) {
            //Set Ethernet switch result
            }
        }

        @Override
        public IBinder asBinder() {
            return super.asBinder();
        }
};
```

Remarks：
-- None

11.2 Unregister the Common listener

Interface name:
int unregisterCommonListener(int appId, ISdkCommonListener listener)

Parameter：

Parameter name	Type	Explain
appId	int	app Type Id Defined in ApiTool.CommonAppId
listener	listener	Listening listener

Returned value：

Parameter name	Type	Explain
Returned value	String	Is the cancellation successful? ApiTool.API_RES_*

Code example:

```
int unregisterCommonListener(ApiTool.CommonAppId.ID_APP_CUSTOM,
listener);
```

Remarks：
-- None

11.3 Set Common Execution Events

Interface name:
int setCommonEvent(int appId, int actionId, String strParam, int[] dataArray, int arrayLen)

Parameter：

Parameter name	Type	Explain
appId	int	app Type Id Defined in ApiTool.CommonAppId
actionId	int	event Id Define in ApiTool.ScaleEvent
strParam	String	The parameter can be null
dataArray	int[]	parameter If not used, initialize it as an empty array. Never assign it to null.
arrayLen	int	Data array length

Returned value：

Parameter name	Type	Explain
Returned value	String	The event result is notified to the listener. For details, see Section 11.4 of the ISDKCommonListener interface documentation.

Code example:

```
int setCommonEvent(ApiTool.CommonAppId.ID_APP_CUSTOM,
ApiTool.CommonEvent.EVENT_GET_LN);
```

Remarks：
-- None

11.4 ISdkCommonListener Interface Description

Interface name:
ISdkCommonListener

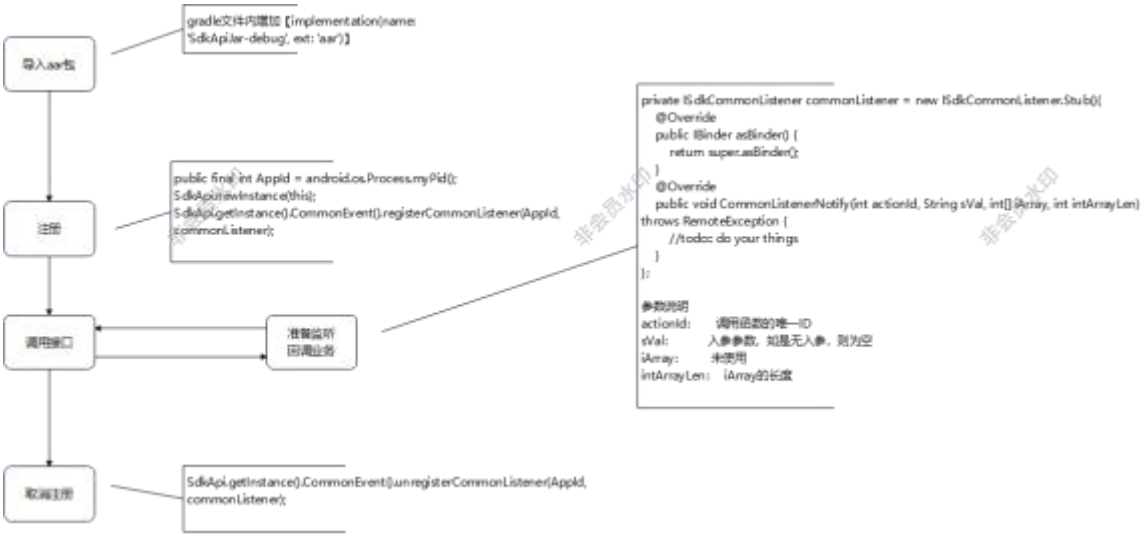
Parameter：

Parameter name	Type	Explain
actionId	int	event Id Defined in ApiTool.CommonEvent and
sVal	String	The interface returns a parameter or the execution result.
iArray	int[]	This parameter typically has no special meaning. The reserved return array may assist the intArrayLen parameter in returning the system interfaces operation results.
intArrayLen	int	The iArray data length. The received length indicates the success/failure status of the interface callback, including error codes in case of failure.

Returned value：
-- None
Code example:
-- None

Remarks：
The callback is triggered after executing setCommonEvent, enabling interfaces with return values to be easily handled.

11.5 CommonEvent Usage



11.6 Interface Call Usage Example

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
actionId, params, dataArray, dataArray.length());
```

parameter declaration：

--AppId: The process ID of the calling app

--actionId: The unique ID for the function call

--params: Input parameters. If multiple parameters are specified, the json parameter is a JSON string.

--dataArray: Backup parameter. If not used, initialize it as an empty array. Never set it to null.

dataArray.length()

11.7 actionId command set

command name	command code	Explain	remarks
EVENT_GET_LT_VERSION	0x1003	Get LongXun chip version	
SWITCH_WIFI	0x9000	WiFi functional switch	
GET_WIFI_MAC	0x9001	Get WiFi MAC	
SWITCH_BLUETOOTH	0x9010	Bluetooth function switch	
GET_BLUETOOTH_MAC	0x9011	Get Bluetooth MAC	
SWITCH_ETH	0x9020	Ethernet function switch	
SWITCH_AP	0x9030	AP functional switch	
GET_ETH_STATUS	0x9040	Get Ethernet status	
SET_ETH_IP	0x9050	Set the Ethernet IP	
GET_ETH_INFO	0x9060	Get network information	
GET_AP_INFO	0x9070	Get AP information	
SET_SCREEN_SATURATION	0x9080	Set screen saturation	
GET_SCREEN_SATURATION	0x9081	Get screen saturation	
SET_SCREEN_CONTRAST	0x9082	Set screen contrast	
GET_SCREEN_CONTRAST	0x9083	Get screen contrast	
SET_SCREEN_HUE	0x9084	Set screen tone	
GET_SCREEN_HUE	0x9085	Get screen tone	

SET_SCREEN_SIGNAL_BRIGHTNESS	0x9086	Set screen brightness	
GET_SCREEN_SIGNAL_BRIGHTNESS	0x9087	Get screen brightness	
UPDATE_SYSTEM_LOGO	0x9090	Update system startup logo	Old plan
UPDATE_SYSTEM_OTA	0x9091	OTA upgrade	
UPDATE_SYSTEM_UBOOT_LOGO	0x9092	Update U-Boot logo	Old plan
UPDATE_SYSTEM_KERNEL_LOGO	0x9093	Update kernel logo	Old plan
UPDATE_SYSTEM_UBOOT_LOGO1	0x9094	Update U-Boot logo	New plan
UPDATE_SYSTEM_KERNEL_LOGO1	0x9095	Update kernel logo	New plan
UPDATE_SYSTEM_ANIMATION	0x9096	Update startup animation	New plan
UPDATE_SYSTEM_WALLPAPER	0x9097	Update wallpaper	New plan
EVENT_CONFIG_WIEGAND_DATA	0x10004	Wagen configuration	
EVENT_SEND_WIEGAND_DATA	0x10005	Send data to Vega	
EVENT_READ_WIEGAND_DATA	0x10006	Wagen receives data	

11.7.1 WiFi function switch

command name：
SWITCH_WIFI（0x9000）

Parameter：

Parameter name	Type	Explain
params	String	on： open off： close

Returned value：
-- None

Code example:

```
// open WiFi
int[] dataArray = new int[]{ 0};
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
SWITCH_WIFI, "on", dataArray, dataArray.length());
```

Remarks：
-- None

11.7.2 Get WiFi MAC

command name：
GET_WIFI_MAC（0x9001）

Parameter：

Parameter name	Type	Explain
params	String	None

Returned value：
-- None

Code example:

```
    Get WiFi MAC
    int[] dataArray = new int[]{0};
    SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
    GET_WIFI_MAC, params, dataArray, dataArray.length());
```

Remarks：
-- None

11.7.3 Bluetooth function switch

command name：
SWITCH_BLUETOOTH（0x9010）

Parameter：

Parameter name	Type	Explain
params	String	on： open off： close

Returned value：
-- None

Code example:

```
Open Bluetooth
int[] dataArray = new int[]{0};
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
GET_BLUETOOTH_MAC , "on", dataArray, dataArray.length());
```

Remarks：
-- None

11.7.4 Ethernet Function Switch

command name：
SWITCH_ETH（0x9020）

Parameter：

Parameter name	Type	Explain

params	String	Option 1:
		on： open
		off： close
		
		Option 2:
		true： open
		false： close
		
		Option 3:
		json class{
		String ethNum; //Ethernet card name, such as eth0
		bool flagSwitch; //Function switch
		}

Returned value：
-- None

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
SWITCH_ETH, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.5 AP Function Switch

command name：

SWITCH_AP（0x9030）

Parameter：

Parameter name	Type	Explain
params	String	See the class structure below When using, convert class to JSON string

The class structure of the parameters

```
classApInfo{
bool switchONOff; //Function switch
    String apName; //AP name
bool apIp; // specifies the IP
String password; //AP password
    boolean offWhenNoDeviceConn;
    /*
*Security Type
    *SECURITY_Type_OPEN = 0
    *SECURITY_Type_WPA2_PSK = 1
    *SECURITY_Type_WPA3_SAE_TRANSITION = 2
    *SECURITY_Type_WPA3_SAE = 3
    */
    private int pwType;
    /*
* tape width
    *BAND_2GHZ = 1 << 0
    *BAND_5GHZ = 1 << 1
    *BAND_6GHZ = 1 << 2
    *BAND_60GHZ = 1 << 3
    */
    private int band;
}
```

Returned value：
-- None

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
SWITCH_AP, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.6 Get Ethernet Status

command name：

GET_ETH_STATUS（0x9040）

Parameter：

Parameter name	Type	Explain
params	String	Network drive name: iface Example: eth0

Returned value：

Parameter name	Type	Explain
sVal	String	true： open false： close

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
GET_ETH_STATUS, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.7 Set Ethernet IP

command name：

SET_ETH_IP（0x9050）

Parameter：

Parameter name	Type	Explain
params	String	See the class structure below When using, convert class to JSON string

The class structure of the parameters

```
class IpInfo{
String_iface; //Network interface name, such as eth0
bool isStatic; //true: static IP mode false: DHCP
    String ipAddr;
    String mask;
    String gate;
    String dns1;
    String dns2;
}
```

Returned value：

Parameter name	Type	Explain
sVal	String	0: Success Other values: Failed

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
SET_ETH_IP, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.8 Access Network Information

command name：

GET_ETH_INFO（0x9060）

Parameter：

Parameter name	Type	Explain
params	String	Network drive name: iface Example: eth0 See the Network Drive Type Parameter Table

Returned value：

Parameter name	Type	Explain
sVal	String	Network drive information

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
GET_ETH_INFO, params, dataArray, dataArray.length());
```

Remarks：
-- None

11.7.9 Get AP information

command name：
GET_AP_INFO（0x9070）

Parameter：

Parameter name	Type	Explain
params	String	null

Returned value：

Parameter name	Type	Explain
sVal	String	AP Drive Information

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
GET_AP_INFO, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.10 Set screen saturation

command name：

SET_SCREEN_SATURATION（0x9080）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	The range of the [0] data array value is 0 to 100.

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
SET_SCREEN_SATURATION, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.11 Get screen saturation

command name：

GET_SCREEN_SATURATION（0x9080）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	null

Returned value：

Parameter name	Type	Explain
sVal	String	Value range: 0-100
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
GET_SCREEN_SATURATION, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.12 Set screen contrast

command name：

SET_SCREEN_CONTRAST（0x9082）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	The range of the [0] data array value is 0 to 100.

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
SET_SCREEN_CONTRAST, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.13 Get screen contrast

command name：

GET_SCREEN_CONTRAST（0x9083）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	null

Returned value：

Parameter name	Type	Explain
sVal	String	Value range: 0-100
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
GET_SCREEN_CONTRAST, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.14 Set screen tone

command name：

SET_SCREEN_HUE（0x9084）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	dataArray[0]：0 - 100

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
SET_SCREEN_HUE, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.15 Get screen tones

command name：

GET_SCREEN_HUE（0x9085）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	null

Returned value：

Parameter name	Type	Explain
sVal	String	Value range: 0-100
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
GET_SCREEN_HUE, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.16 Update the system startup logo

command name：

UPDATE_SYSTEM_LOGO（0x9090）

Parameter：

Parameter name	Type	Explain
params	String	See the class structure below When using, convert class to JSON string

The class structure of the parameters

```
classLogoInfo{
    int Type;  //0:animation  1:boot logo
    String orientation; //horizontal: landscape vertical:
portrait
    String logoFilePath; //Absolute path to the file
    int logoFileType;  //0:png  1:zip  2:mp4
}
```

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]: 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
UPDATE_SYSTEM_LOGO, params, dataArray, dataArray.length());
```

Remarks：

This interface is being phased out. Use 0x9094 instead.

11.7.17 OTA update

command name：

UPDATE_SYSTEM_OTA（0x9091）

Parameter：

Parameter name	Type	Explain
params	String	Absolute path of the OTA file The default path is /mnt/sdcard/update.zip

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
UPDATE_SYSTEM_OTA, params, dataArray, dataArray.length());
```

Remarks：

Before upgrading, ensure there is enough local storage space.

11.7.18 Update the u-boot logo

command name：

UPDATE_SYSTEM_UBOOT_LOGO （0x9092）

Parameter：

Parameter name	Type	Explain
params	String	Absolute path to the Uboot logo file The file must be in BMP format. --The BMP image must be 8-bit or 24-bit The image sizes and resolutions of boot.bmp and boot kernel.bmp must be identical.

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
UPDATE_SYSTEM_UBOOT_LOGO, params, dataArray, dataArray.length());
```

Remarks：

This interface is being phased out. Use 0x9094 instead.

11.7.19 Update the kernel logo

command name：

UPDATE_SYSTEM_KERNEL_LOGO （0x9093）

Parameter：

Parameter name	Type	Explain
params	String	absolute path of the kernel.logo file The file must be in BMP format. --The BMP image must be 8-bit or 24-bit The image sizes and resolutions of boot.bmp and boot kernel.bmp must be identical.

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
UPDATE_SYSTEM_KERNEL_LOGO, params, dataArray,
dataArray.length());
```

Remarks：

This interface is being phased out. Use 0x9095 instead.

11.7.20 Update the u-boot logo

command name：

UPDATE_SYSTEM_UBOOT_LOGO1（0x9094）

Parameter：

Parameter name	Type	Explain
params	String	See the class structure below When using, convert class to JSON string

The class structure of the parameters

```
classLogoInfo{
    int Type;    //0:animation  1:logo  2:wallpaper
String orientation; //horizontal: landscape vertical: portrait
String file path; //Absolute path to the file
    int fileType;        //0:image 1:zip  2:video
int actionType; //0: Import 1: Clear imported files
}

pay attention to :
--logo only supports
  >> BMP format file
  >> Depth 24
  The maximum resolution is 4K, subject to the actual
capabilities of the board.
```

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]: 0: Success 1: Failed 2: The interface is not supported iArray[1]: Read only when iArray[0] = -1 --See the detailed error code in **LOGO file import error code table** ”

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
UPDATE_SYSTEM_KERNEL_LOGO1, params, dataArray,
dataArray.length());
```

Remarks：
-- None

11.7.21 Update the kernel logo

command name：
UPDATE_SYSTEM_KERNEL_LOGO1 （0x9095）

Parameter：

Parameter name	Type	Explain
params	String	See the class structure below When using, convert class to JSON string

The class structure of the parameters

```
class LogoInfo{
    int Type;    //0:animation  1:logo  2:wallpaper
String orientation; //horizontal: landscape vertical: portrait
String file path; //Absolute path to the file
    int fileType;        //0:image 1:zip  2:video
int actionType; //0: Import 1: Clear imported files
}

pay attention to :
--logo only supports
>> BMP format file
>> Depth 24
The maximum resolution is 4K, subject to the actual capabilities
of the board
```

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported iArray[1]: Read only when iArray[0] = -1 --See the detailed error code in **LOGO file import error code table** ”

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
UPDATE_SYSTEM_KERNEL_LOGO1, params, dataArray,
dataArray.length());
```

Remarks：
-- None

11.7.22 Update startup animation

command name：
UPDATE_SYSTEM_ANIMATION（0x9096）

Parameter：

Parameter name	Type	Explain
params	String	See the class structure below When using, convert class to JSON string

The class structure of the parameters

```
class LogoInfo{
    int Type;    //0:animation  1:logo  2:wallpaper
String orientation; //horizontal: landscape vertical: portrait
String file path; //Absolute path to the file
    int fileType;        //0:image 1:zip  2:video
int actionType; //0: Import 1: Clear imported files
}

pay attention to :
--logo only supports
  zip and mp4 [h264 or h265] files
  >> Maximum file size 60MB
  The maximum resolution is 4K, subject to the actual
capabilities of the board.
```

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported iArray[1]: Read only when iArray[0] = -1 --See the detailed error code in **LOGO file import error code table** ”

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
UPDATE_SYSTEM_ANIMATION, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.23 Update wallpaper

command name：

UPDATE_SYSTEM_WALLPAPER（0x9097）

Parameter：

Parameter name	Type	Explain
params	String	See the class structure below When using, convert class to JSON string

The class structure of the parameters

```
class LogoInfo{
    int Type;    //0:animation  1:logo  2:wallpaper
String orientation; //horizontal: landscape vertical: portrait
String file path; //Absolute path to the file
    int fileType;        //0:image 1:zip  2:video
int actionType; //0: Import 1: Clear imported files
}

pay attention to :
--logo only supports
 PNG format file
 >> Maximum file size is 10MB
 The maximum resolution is 5000x5000, subject to the actual
capabilities of the board
```

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported iArray[1]: Read only when iArray[0] = -1 --See the detailed error code in **LOGO file import error code table** ”

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
UPDATE_SYSTEM_ WALLPAPER, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.24 Set screen brightness

command name：

SET_SCREEN_SIGNAL_BRIGHTNESS（0x9086）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	The DataArray[0] value ranges from 0 to 100

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
SET_SCREEN_SIGNAL_BRIGHTNESS, params, dataArray,
dataArray.length());
```

Remarks：
-- None

11.7.25 Get screen signal brightness

command name：
GET_SCREEN_SIGNAL_BRIGHTNESS（0x9087）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	null

Returned value：

Parameter name	Type	Explain
sVal	String	Value range: 0-100
iArray	int[]	iArray[0]: 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
GET_SCREEN_SIGNAL_BRIGHTNESS , params, dataArray,
dataArray.length());
```

Remarks：
-- None

11.7.26 Wagen Configuration

command name：
EVENT_CONFIG_WIEGAND（0x10004）

Parameter：

Parameter name	Type	Explain
params	String	Wagen bit, content is “26”、“32”、“34”、“58”
dataArray	int[]	You need to initialize 1-byte data with content 0

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
EVENT_CONFIG_WIEGAND, params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.27 Wagen sends data

command name：

EVENT_SEND_WIEGAND_DATA（0x10005）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	--int array: Transmits fixed-size data based on the Wiegand bit Example: In 26-bit mode, transmit 3-byte data: dataArray[3] = {0x01,0x02,0x03}

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： 0: Success 1: Failed 2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
EVENT_SEND_WIEGAND_DATA , params, dataArray, dataArray.length());
```

Remarks：

-- None

11.7.28 Wagen receives data

command name：

EVENT_READ_WIEGAND_DATA（0x10006）

Parameter：

Parameter name	Type	Explain
dataArray	int[]	--int array: requires initialization of 1-byte data with content 0 Suggestion: Read the Wiegand data every 500ms

Returned value：

Parameter name	Type	Explain
iArray	int[]	iArray[0]： --0: Success >> strVal: Returns the read data; otherwise, returns failure -- -1: Failed -- -2: The interface is not supported

Code example:

```
SdkApi.getInstance().CommonEvent().setCommonEvent(AppId,
EVENT_READ_WIEGAND_DATA , params, dataArray, dataArray.length());
```

Remarks：

-- None

API constant

LOGO file import error code table

variable	Code value	Explain
FILE_NOT_EXIST	0x0001	The file does not exist
FILE_NOT_READABLE	0x0002	The file is not readable
GSON_FORMAT_FAIL	0x0003	String escape failed for JSON
INVALID_GSON_INFO	0x0004	The data does not match the LOGO class definition
INVALID_ORIENTATION	0x0005	Direction parameter field does not match
INVALID_SOURCE_FILE_PATH	0x0006	The path parameter field is empty
NO_ENOUGH_CACHE_SIZE	0x0007	The target directory has insufficient storage space
INVALID_FILE_SUFFIX	0x0008	Invalid file extension
FILE_COPY_FAIL	0x0009	File copy failed
FILE_SIZE_OVER_LIMIT	0x000A	File size exceeds the limit
BMP_HEADER_SIZE_INVALID	0x1010	The BOOT image file header size is incorrect
BMP_SIGNATURE_INVALID	0x1012	BOOT image verification failed
BMP_FILE_SIZE_NOT_MATCH	0x1013	The BOOT image file size does not match
BMP_BIT_DEPTH_UNSUPPORTED	0x1014	BOOT image depth is not supported
BMP_DECODE_FAIL	0x1015	BOOT image decoding failed
BMP_RESOLUTION_NOT_SUPPORT	0x1016	The BOOT image resolution is not supported

ANIMATION_MP4_SUFFIX_INVALID	0x1020	MP4 animation files are not supported
ANIMATION_MP4_HEADER_INVALID	0x1021	The MP4 animation file header format is incorrect
ANIMATION_MP4_DECODE_FAIL	0x1022	MP4 animation file decoding failed
ANIMATION_MP4_FORMAT_UNSUPPORTED	0x1023	MP4 animation file verification failed
ANIMATION_ZIP_FORMAT_NOT_STORED_ONLY	0x1024	The ZIP animation file is not stored in compressed format.
ANIMATION_ZIP_DESC_NOT_FOUND	0x1025	The description file in the ZIP animation file is not found.
ANIMATION_ZIP_DESC_INVALID	0x1026	The ZIP animation file contains an invalid description file.
ANIMATION_ZIP_PNG_INVALID	0x1027	The ZIP animation file contains PNG files.
ANIMATION_MP4_RESOLUTION_NOT_SUPPORT	0x102A	MP4 animation file resolution is not supported
ANIMATION_ZIP_DECODE_FAIL	0x102B	ZIP animation file decoding failed
WALLPAPER_DECODE_FAIL	0x1030	Wallpaper image decoding failed
WALLPAPER_FORMAT_UNSUPPORTED	0x1031	The wallpaper image format is not supported
WALLPAPER_RESOLUTION_NOT_SUPPORT	0x1032	The wallpaper image resolution is not supported

Network driver Type parameter table

Variable	Code value	Description
NET_Type_ETH0	eth0	Ethernet Network Card 1
NET_Type_ETH1	eth1	Ethernet Network Card 2
NET_Type_WIFI	wifi	WiFi
NET_Type_MOBILE	mobile	Mobile network

GPIO Value Definition

Group number is defined as follows

```
    public static final int GPIO0 = 0;
    public static final int GPIO1 = 1;
    public static final int GPIO2 = 2;
    public static final int GPIO3 = 3;
    public static final int GPIO4 = 4;
    public static final int GPIO5 = 5;
    public static final int GPIO6 = 6;
    public static final int GPIO7 = 7;
    public static final int GPIO8 = 8;
```

The IO sequence number is defined as follows

```
    public static final int PA0 = 0;
    public static final int PA1 = 1;
    public static final int PA2 = 2;
    public static final int PA3 = 3;
    public static final int PA4 = 4;
    public static final int PA5 = 5;
    public static final int PA6 = 6;
    public static final int PA7 = 7;

    public static final int PB0 = 8;
    public static final int PB1 = 9;
    public static final int PB2 = 10;
    public static final int PB3 = 11;
    public static final int PB4 = 12;
    public static final int PB5 = 13;
    public static final int PB6 = 14;
    public static final int PB7 = 15;

    public static final int PC0 = 16;
    public static final int PC1 = 17;
    public static final int PC2 = 18;
    public static final int PC3 = 19;
    public static final int PC4 = 20;
    public static final int PC5 = 21;
    public static final int PC6 = 22;
    public static final int PC7 = 23;

    public static final int PD0 = 24;
    public static final int PD1 = 25;
    public static final int PD2 = 26;
    public static final int PD3 = 27;
    public static final int PD4 = 28;
    public static final int PD5 = 29;
    public static final int PD6 = 30;
    public static final int PD7 = 31;
```